Topic 14

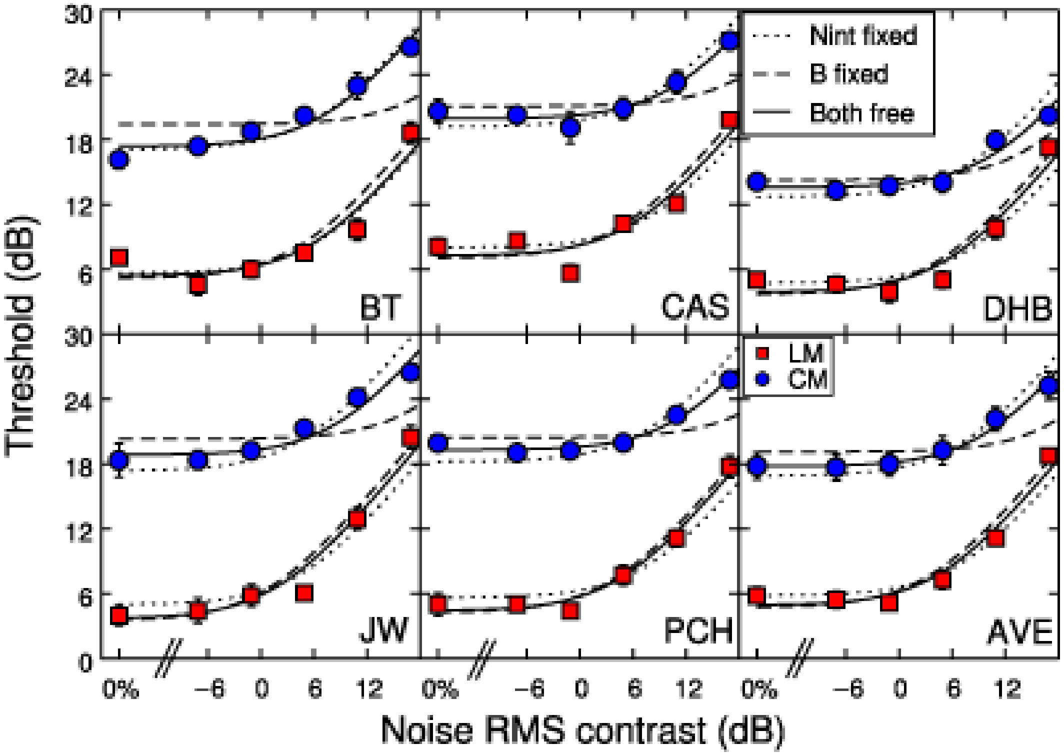
Small-N and quasi-experimental designs

Overview

- Small-N studies
- Quasi-experimental designs
- Questions

Small-N studies

- Psychophysics

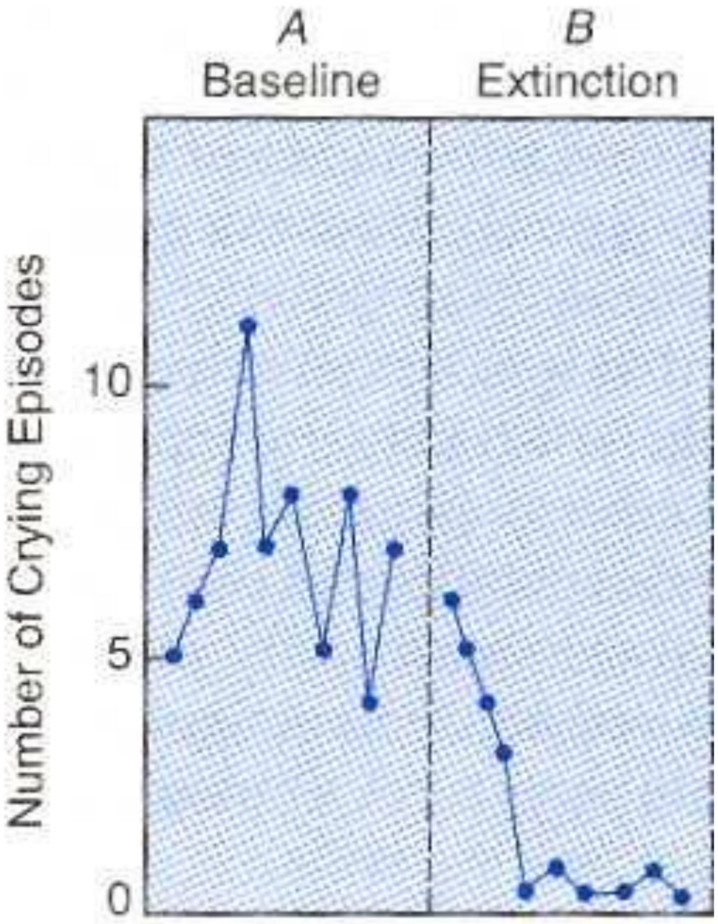


- Psychophysical study
- N = 5
- Participants: BT, CAS, DHB, JW, PCH
- AVE = average
- *"if you've seen one, you've seen them all"*

Small-N studies

- Psychophysics
- Behavior-change studies
  - Stable-baseline design (AB Design)
    - A = behavior during baseline
      B = behavior during treatment / after treatment

Small-N studies



Small-N studies

- Psychophysics
- Behavior-change studies
  - Stable-baseline design (AB Design)
    - A = behavior during baseline
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  - Reversal design (ABA or ABAB Design)

Small-N studies

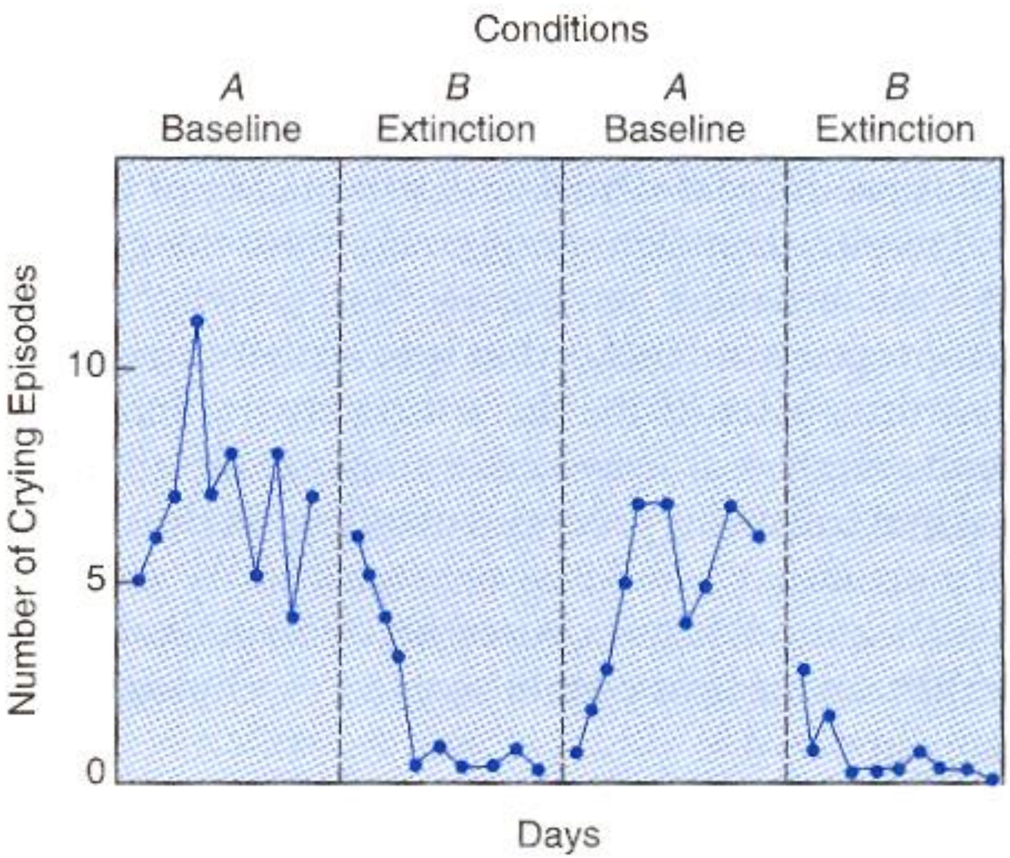

Figure 11–1 The number of crying episodes exhibited by Bill, a nursery-school student, during the four phases of an *ABAB* design initiated to control his problem crying (From Wolf & Risley, 1969, p. 316).

Small-N studies

- Psychophysics
- Behavior-change studies
  - Stable-baseline design (AB Design)
    - A = behavior during baseline
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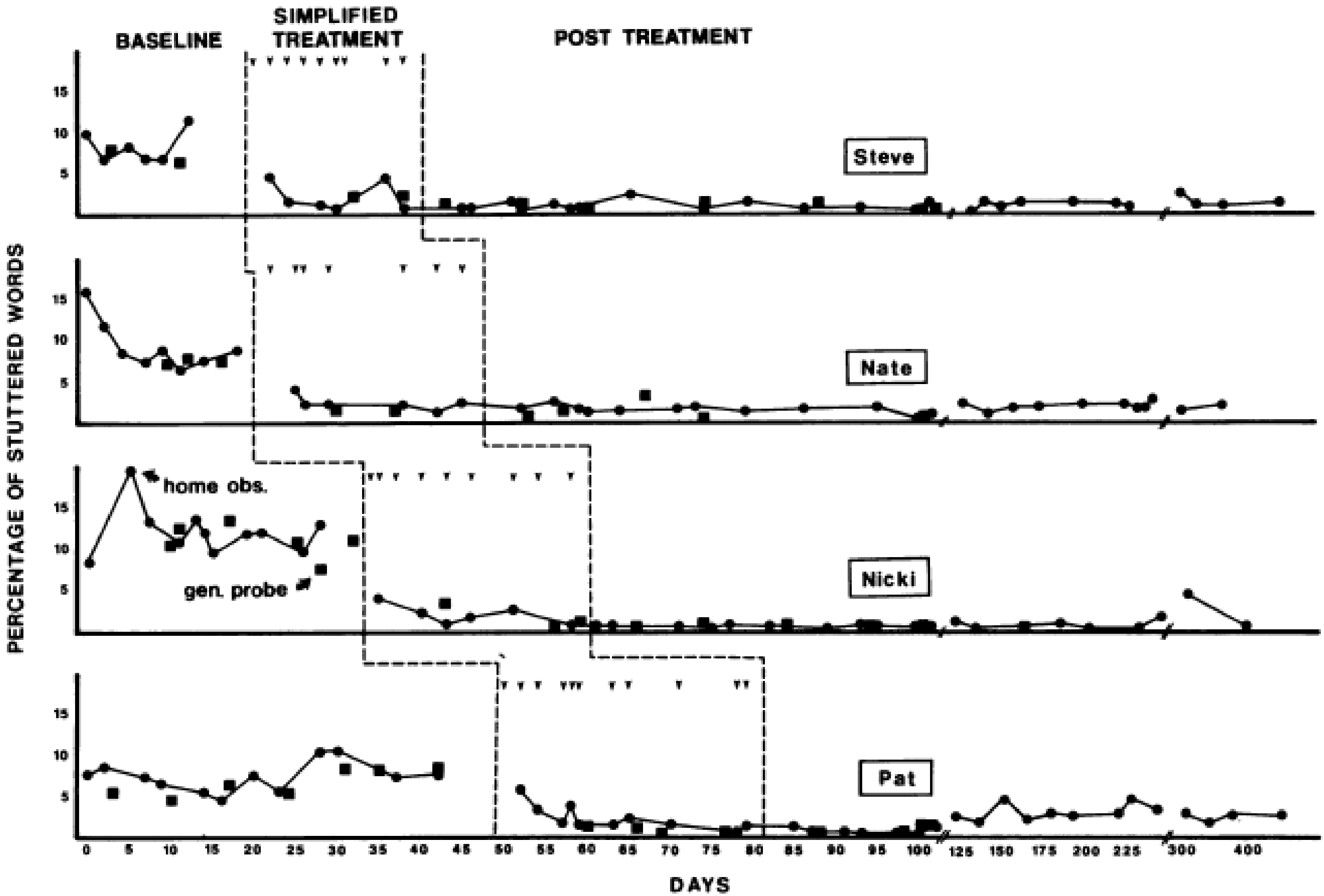
  - Reversal design (ABA or ABAB Design)
  - Multiple-baseline design

Small-N studies



Small-N studies

- Psychophysics
- Behavior-change studies
  - Stable-baseline design (AB Design)
    - A = behavior during baseline
      B = behavior during treatment / after treatment
  - Reversal design (ABA or ABAB Design)
  - Multiple-baseline design
- Case studies

Small-N studies

- Small-N studies → often aim to generate causal claims
  - Internal validity is often not high.
  - It is also important to question how well
    - each variable was manipulated and measured (construct validity)
    - the results generalize (external validity)
    - the numbers support the conclusions (statistical validity)

Quasi-experimental designs

- Experiments in which the experimenter selects rather than manipulates the independent variables.
  - Environmental variables
  - Subject variables

Quasi-experimental designs

- Pretest/posttest design (AB design)
  O_1 - T - O_2
  - e.g., "study on the effectiveness of flower essence therapy on depression"

This article presents the findings of a series of studies conducted to determine the clinical efficacy of flower essences on the treatment of mild to moderate depression. Funding for the study was provided by the Flower Essence Society.

http://www.flowersociety.org/depression-study.pdf

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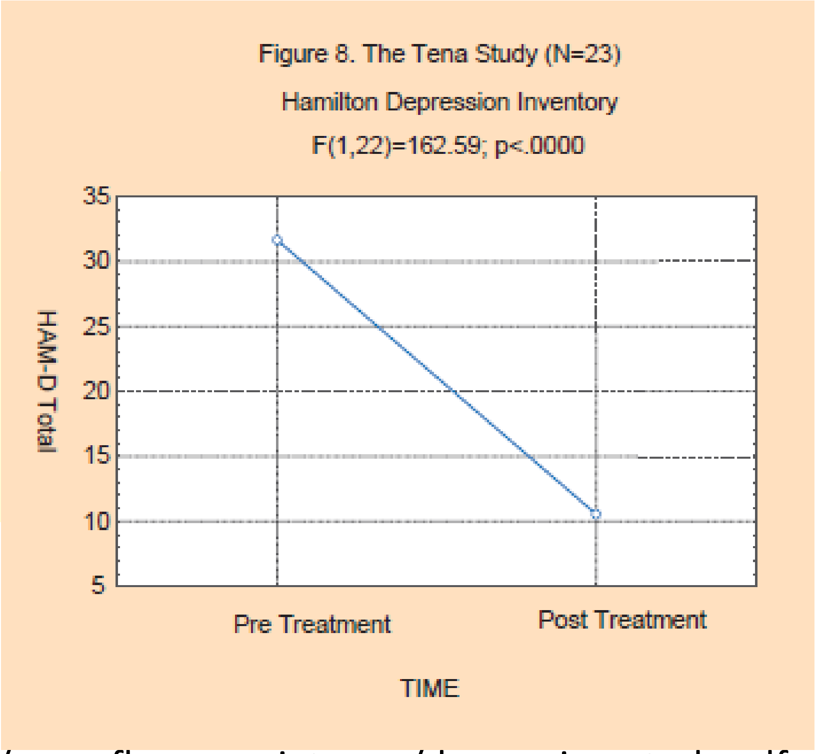

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Quasi-experimental designs

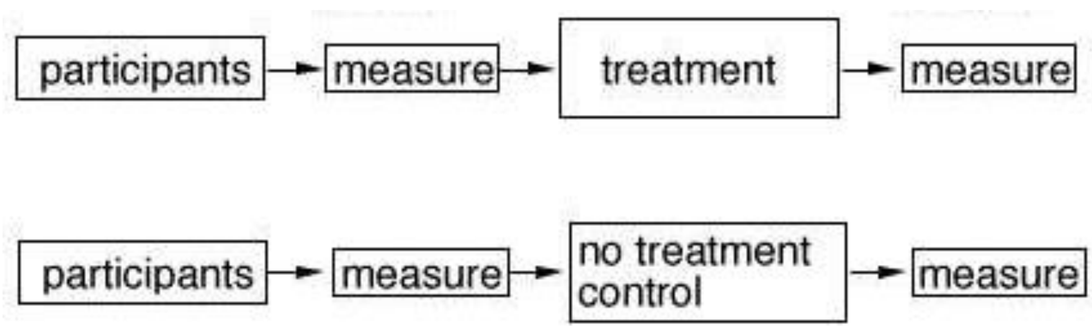
- Pretest/posttest design (AB design)

  O_1 - T - O_2
- Nonequivalent control group pretest/posttest design

  O_1 - T - O_2 experimental group (E)

  O_1 - - O_2 nonequivalent control group (C)

  - Matching

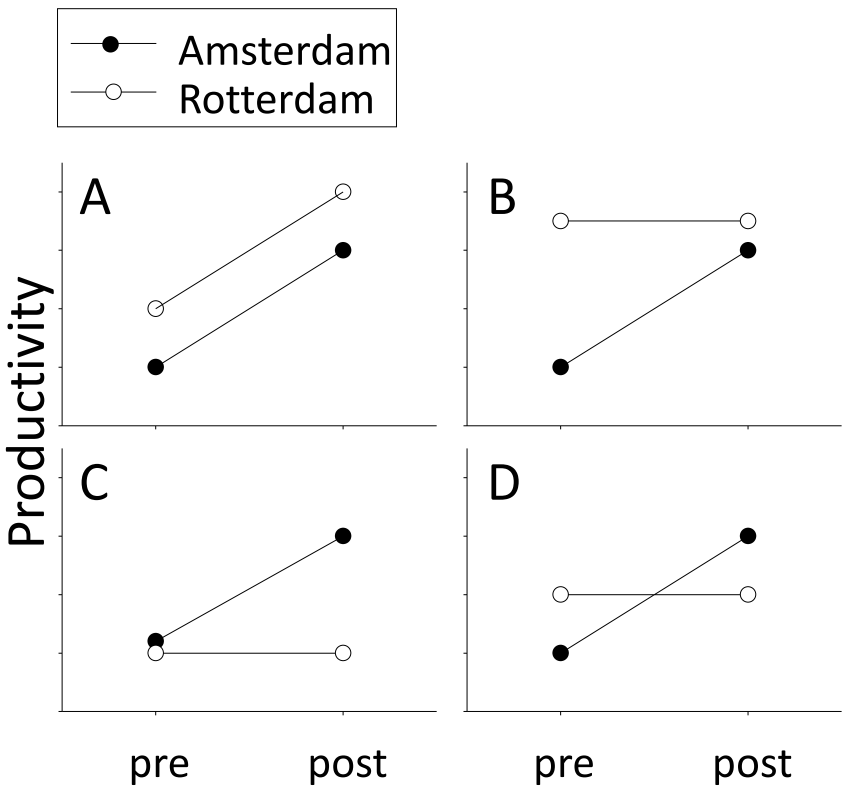


Quasi-experimental designs

- An example of a nonequivalent control group pretest/posttest design
  - Is there a change in productivity after the introduction of a flex working schedule?
    - E = branch in Amsterdam
    - C = branch in Rotterdam

Quasi-experimental designs

- An example of a nonequivalent control group pretest/posttest design
  - Is there a change in productivity after the introduction of a flex working schedule?



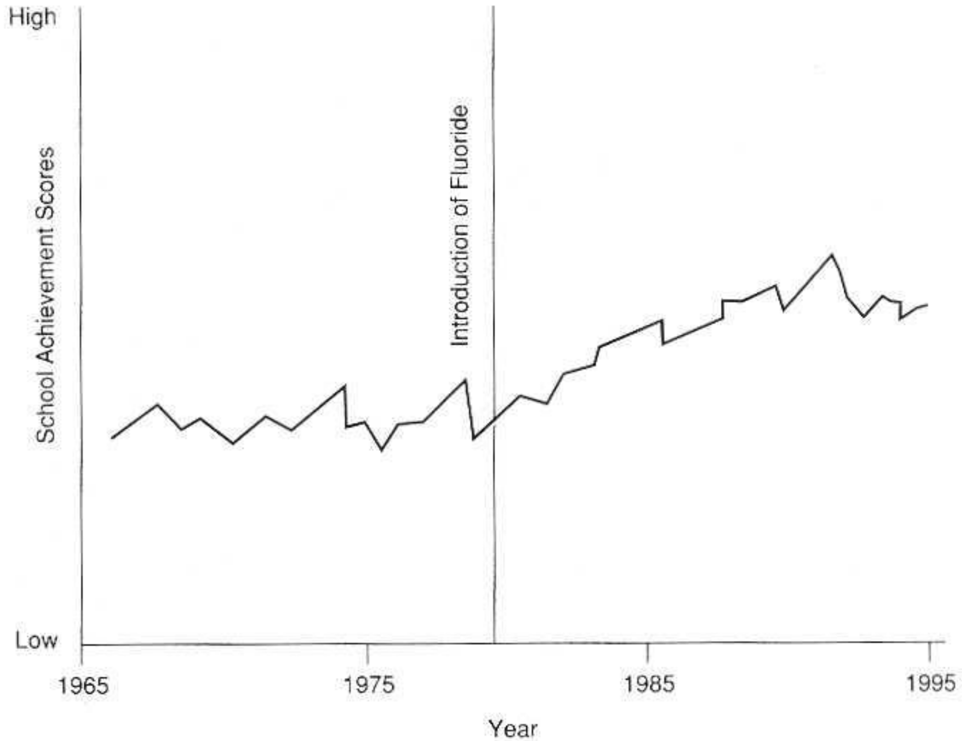
A.
Equal change in E and C: no evidence for effect of T
B.
possible ceiling effect Rotterdam: no good evidence for effect of T
C & D.
Good evidence for effect of T (but beware of Hawthorne effect!)

Quasi-experimental designs

- Pretest/posttest design (AB design)
  O_1 - T - O_2
- Nonequivalent control group pretest/posttest design
  O_1 - T - O_2 experimental group (E)
  O_1 - - O_2 nonequivalent control group (C)
- Interrupted time-series design
  O_1 O_2 O_3 O_4 O_5 T O_6 O_7 O_8 O_9 O_{10}

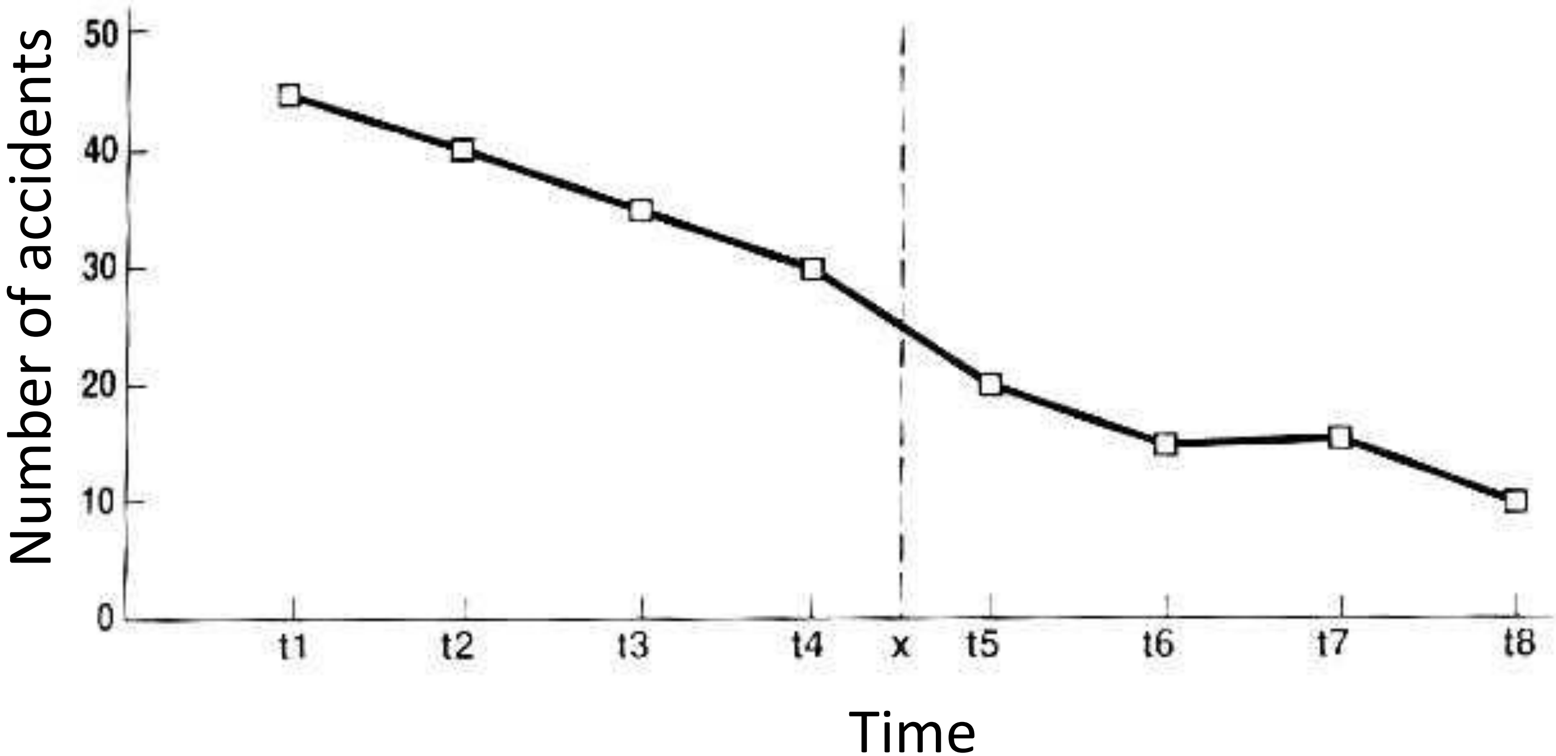
Quasi-experimental designs

- An example of an interrupted time-series design
  - Effect of fluoride in drinking water on school performance



Quasi-experimental designs

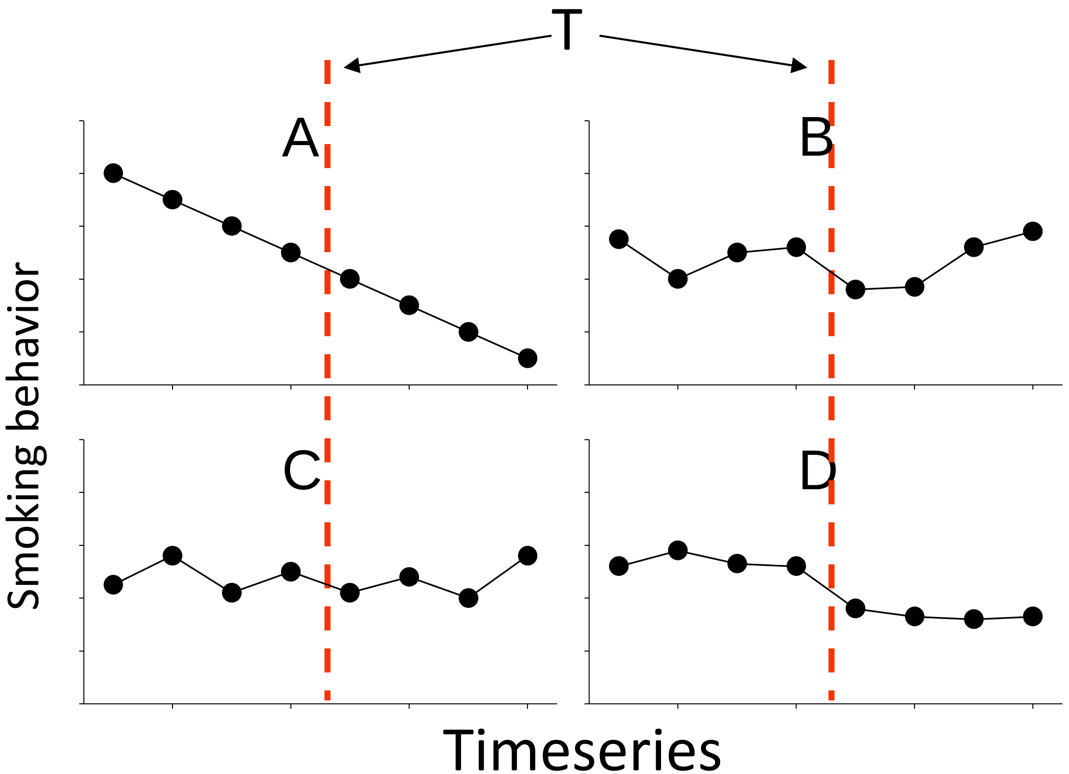
- An example of an interrupted time-series design
  - Effect of new traffic measurement on number of accidents



X = Introduction of new traffic measurement

Quasi-experimental designs

- An example of an interrupted time-series design
  - Effect anti-smoking campaign on smoking behavior in teens



A.

Continuation of linear trend: No evidence for effect of T.

B.

Maybe temporary effect of T.

C.

Continuation of periodic fluctuations: no evidence for effect of T.

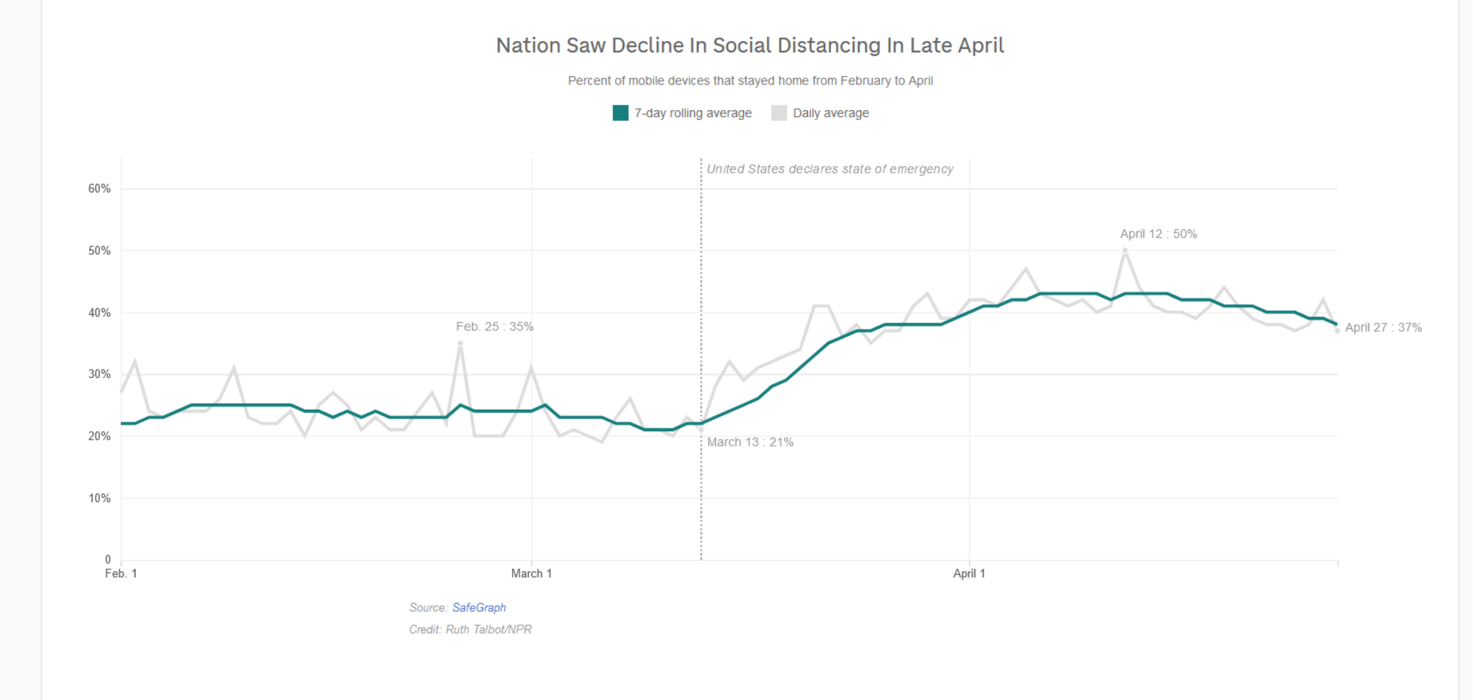
D.

Best possible evidence for effect of T.

Quasi-Experimental Questions

From spring 2020: Are people staying home because of the lockdown? Cell phone data reveal they did, at first.

Story: https://n.pr/2B0nCBm



Quasi-experimental designs

- Pretest/posttest design (AB design)
  O_1 - T - O_2
- Nonequivalent control group pretest/posttest design
  O_1 - T - O_2 experimental group (E)
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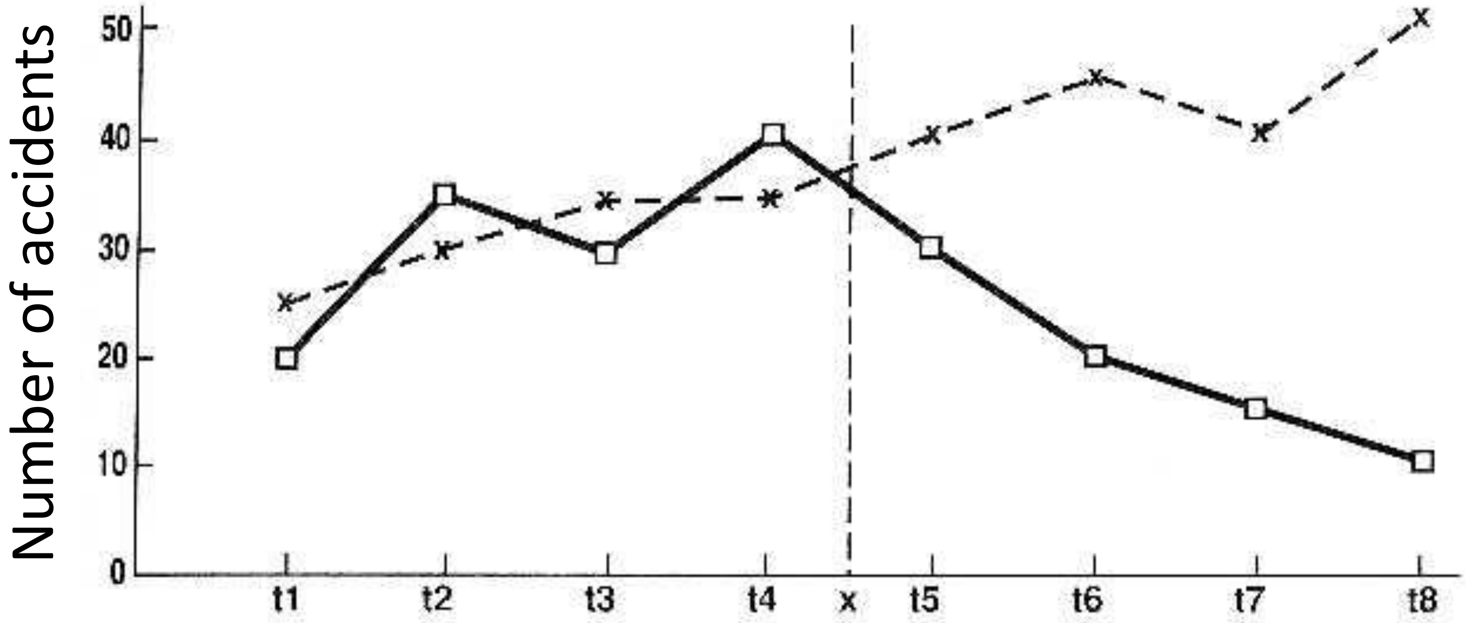
- Interrupted time-series design
  $O_1\ O_2\ O_3\ O_4\ O_5\ T\ O_6\ O_7\ O_8\ O_9\ O_{10}$
- Nonequivalent control group interrupted time-series design
  $O_1\ O_2\ O_3\ O_4\ O_5\ T\ O_6\ O_7\ O_8\ O_9\ O_{10}$
  $O_1\ O_2\ O_3\ O_4\ O_5\ \ O_6\ O_7\ O_8\ O_9\ O_{10}$

Quasi-experimental designs

- An example of a nonequivalent control group interrupted time-series design
  - Effect replacement of crossing by roundabout



□ Neighborhood in which crossing is replaced by roundabout at time *x*.
X matched neighborhood in which crossing is not replaced.

Quasi-experimental designs

- Pretest/posttest design (AB design)

  O_1 - T - O_2
- Nonequivalent control group pretest/posttest design

  O_1 - T - O_2 experimental group (E)

  O_1 - - O_2 nonequivalent control group (C)
- Interrupted time-series design

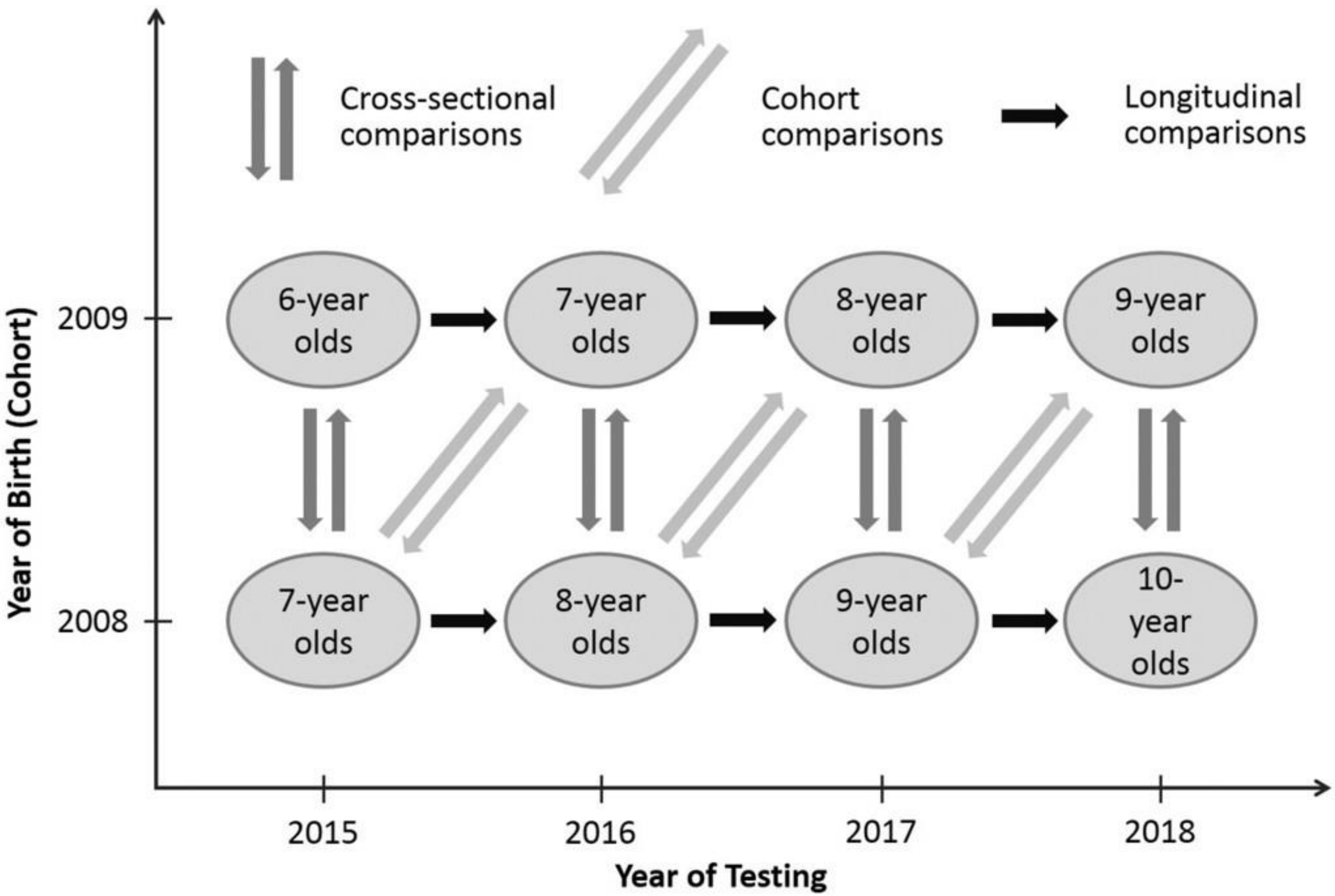
  $O_1 \quad O_2 \quad O_3 \quad O_4 \quad O_5 \quad T \quad O_6 \quad O_7 \quad O_8 \quad O_9 \quad O_{10}$
- Nonequivalent control group interrupted time-series design

  $O_1 \quad O_2 \quad O_3 \quad O_4 \quad O_5 \quad T \quad O_6 \quad O_7 \quad O_8 \quad O_9 \quad O_{10}$

  $O_1 \quad O_2 \quad O_3 \quad O_4 \quad O_5 \quad \quad O_6 \quad O_7 \quad O_8 \quad O_9 \quad O_{10}$
- The subject variable "Age"

Quasi-experimental designs

Cross-Sequential



Quasi-experimental designs

- Quasi-experimental studies → aim to generate causal claims
  - Internal validity is not as high as in a true experiment
  - It is also important to question how well
    - each variable was manipulated and measured (construct validity)
    - the results generalize (external validity)
    - the numbers support the conclusions (statistical validity)

Questions

A politician believes that the use of car lights during the day contributes to greater safety on the road in the Netherlands.

He relies on a literature review written by a major transportation company.

The review reports several studies showing that the number of road accidents decreased significantly in Sweden after the use of car lights during the day became mandatory.

The conclusion of the review was:
Based on the results of previous studies (i.e., those performed in Sweden), the use of car lights during the day in Netherlands should lead to a higher safety on the road.

Why is this conclusion not valid?

A researcher wants to know whether the volume of background noise has an effect on the speed with which people solve calculation assignments.

They perform an experiment in which they manipulate the following independent variable:

(*A*) Volume of background noise (0 dB, 40 dB, and 80 dB).

A constitutes a within-subjects variable.

Six participants take part in the study.

He uses a complete counterbalanced design.

Indicate the order of the conditions for the different participants.

Chess Grandmaster Alexander Aljechin was convinced that the presence of his cat had a positive effect on his chances to win a chess game.

As a result of this belief, he took his cat to each game he had to play.

Suppose that Aljechin took his cat to a chess tournament, and he won all games.

Why would this outcome not provide sufficient justification for the hypothesis that his cat is a win-inducing organism?

An owner of a "health shop" claims that the daily intake of extra vitamins leads to a better heath condition.

He relies on a large study carried out by another health shop owner.

This person recruited subjects for his study in the waiting room of a GP by asking them if they were willing to take extra vitamin pills for a period of 4 months.

All patients complied and used extra vitamin pills for four months.

Fifty subjects took part in the experiment.

Immediately after they were recruited, all participants took part in a test to determine how much time it took them to walk a distance of 1 km.

After four months (and swallowing many extra vitamins) participants took the same test again.

The results:

-average time of the 1st measurement (jan): 10 min

-average time of the 2nd measurement (jun): 7 min

People needed less time to walk 1 km after compared to before the treatment.

Based on these results it was concluded that extra vitamin intake has a positive effect on health condition.

How is this conclusion not valid?

How can we fix the design?

A study includes the following independent variables:
(A) sex (male vs. female) and
(B) use of anabolic steroids (placebo versus anabolic steroids).

The dependent variable is the number of times someone is capable to jump over a fence (for 2 minutes).

The results (averages of 50 subjects):

- male placebo	5.5
- male anabolic steroids	10.5
- female placebo	0.5
- female anabolic steroids	3.5

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- male placebo	5.5
- male anabolic steroids	10.5
- female placebo	0.5
- female anabolic steroids	3.5

The conclusion is threefold:

1) Male participants are better than female participants in jumping.
2) Anabolic steroids improve jumping performance.
3) The effect of anabolic steroids is larger in male participants than in female participants.

Which of these three conclusions is/are not valid and why not?
How can this study be improved?

In a hypothetical study it was shown that there is a positive correlation between the number of times people go to a bar and the amount of alcohol consumption.

A scientist is interested in the question whether going to a bar leads to the consumption of more alcohol or the consumption of more alcohol leads to more bar visits.

Indicate how the scientist might investigate this.

Which result would be in line with the conclusion that going to a bar leads to higher alcohol consumption?

A survey company has just released a new test which was developed to determine someone's decision capability.

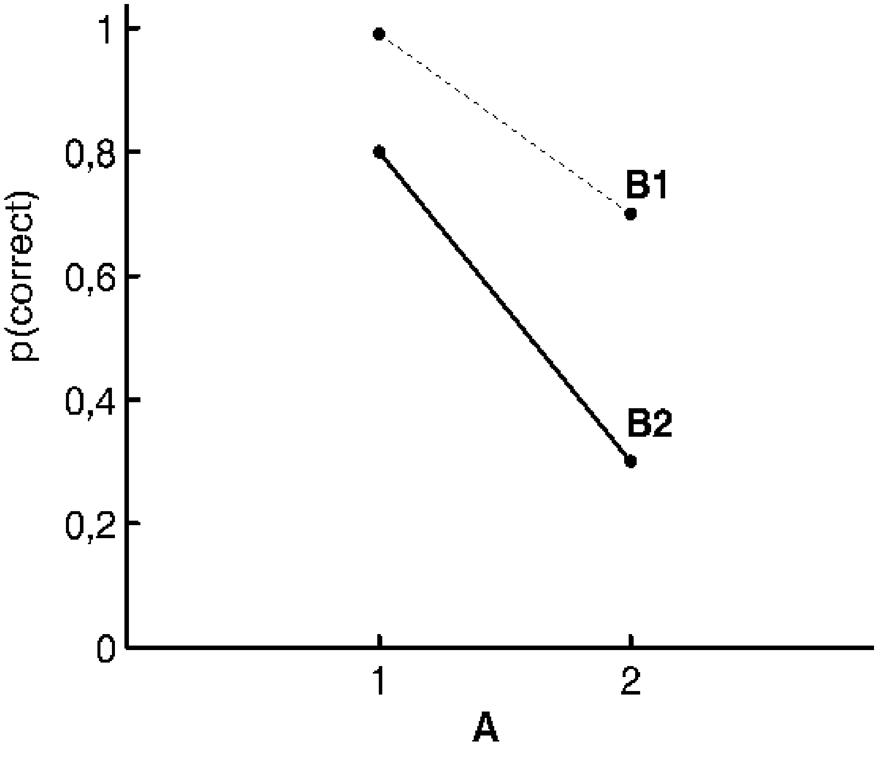
The 2-hour test consisting of 50 items should enable companies to select candidates for high-profile jobs.

A critical researcher likes to know how reliable the test is.

To determine this, he submits 50 participants to the test and calculates the correlation between the scores obtained in one half of the test and those obtained in the other half of the test.

The correlation between the two halves of the test is +.05

What can you conclude about the <u>reliability</u> and the <u>validity</u> of the test?



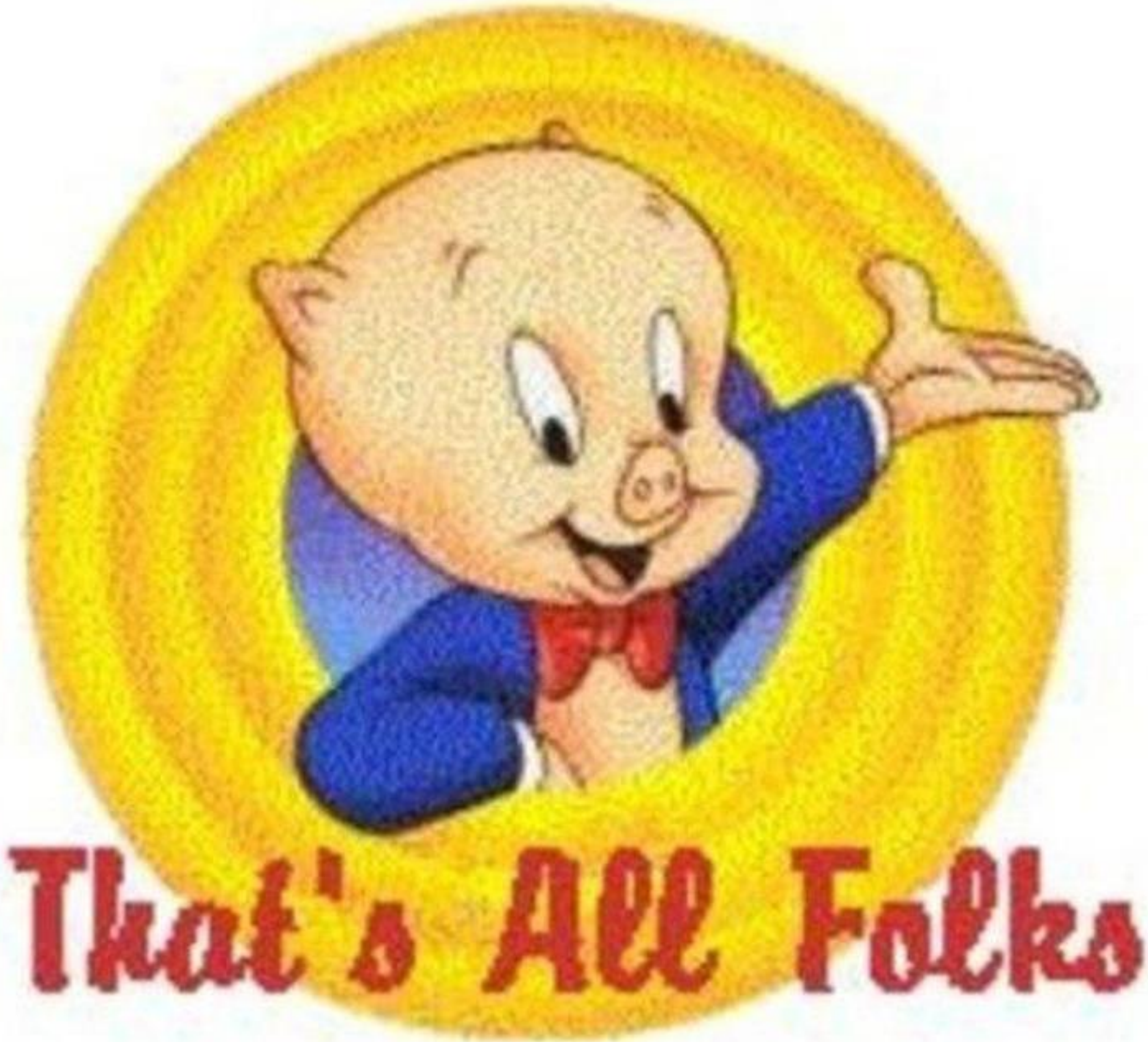
The graph shows the results of a detection experiment.

The statistical analyses show amongst other a significant interaction between A and B.

This implies that the effect of A is different for B1 than for B2.

Is this conclusion valid?

If not, how can the experiment be improved?


That's All Folks